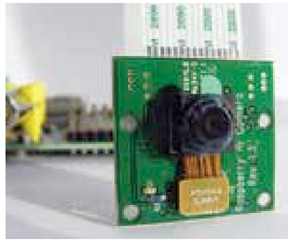
The increased processing power of the Raspberry Pi 2 as compared to the Raspberry Pi model B+ has broadened its range of capabilities. Despite being cheap, it's a powerful piece of hardware in the hands of a person building a working prototype of an idea.

Let's look at some of the possible do-it-yourself applications using the Raspberry Pi.

Detect faces using Raspberry Pi

The added processing power in the Raspberry Pi 2 can be very well exploited by image processing algorithms. Image processing algorithms perform complex calculations on an image to recognise various features in an image, and are among the most computationally expensive algorithms.



Raspberry Pi webcam

How do we do it?

To simplify the process, you can use the PiCam – a camera specially made for use with the Raspberry Pi – using the guide at *http://dgit.in/1HOGcUh.* Or you can connect a USB web camera available in market. For the purpose of this DIY, we'll be using a USB web camera.

To use a USB webcam, you'll need to install the fswebcam package from the Raspberry Pi repository. To install the same, open the terminal and enter the following commands:

- `sudo apt-get update`
- `sudo apt-get install fswebcam`

To check if your webcam works correctly, use the following command:

- `fswebcam image.jpg`

You can specify the resolution of the image using the -r flag along with the required resolution as:

- `fswebcam -r 1280x720 image2.jpg`

Before we continue, we need to install the OpenCV (Open Computer Vision) library that can help detect faces. To install the library, open the terminal and enter the following commands:

- `sudo apt-get install libopencv-dev`